

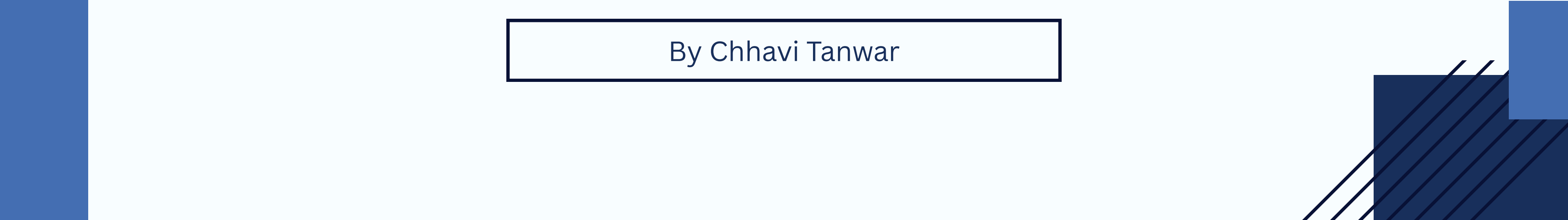


UNDERGRADUATE PROJECT - I

PRESENTATION

**Design and Structural Sizing of an Airbag-Based
Lunar Rover Lander System**

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PROBLEM STATEMENT

The purpose of this project is to create a light but structurally strong landing system for a rover that will explore the lava caves on the Moon. The landing system will include an 8U CubeSat rover (approximately 10 kg in weight), within the total system mass of 15 kg, deployed using an airbag-based landing mechanism inspired by missions like Mars Pathfinder and Mars Exploration Rover. The primary challenge is to design the rigid inner structure that will endure the forces of landing while remaining very light. This includes studying different geometric configurations of the structure, such as the tetrahedron and cube shapes, choosing the right materials, and studying the forces acting on the structure.

EXPECTED OUTCOMES

This project results in the formulation of analytical sizing equations to determine structural dimensions based on impact conditions, material properties, and geometry. It includes evaluation of different structural configurations and materials to optimise mass and strength. A complete conceptual design of the lander can be established, including integration with the descent module, rover payload, and airbag system, along with identification of subsystem interactions. It is also expected to generate CAD models representing both the stowed and deployed configurations of the lander. A key outcome is the comparison of structural designs, showing that tetrahedral structures are more efficient than cubic ones due to axial load transfer, while cubic designs are governed by bending and buckling.



MOTIVATION

The discovery of lunar lava caves is considered to be very promising since they can serve as a shelter against the harmful effects of solar radiation, micrometeorites, and extreme temperature changes. It makes them attractive sites of future habitation and exploration activities on the Moon. But to explore these places, one should have efficient robots with the ability to land safely on the surface. Traditional landing systems are often complex and mass-intensive, making them unsuitable for small-scale missions. Airbag landing systems offer a practical alternative, being lightweight and having been successfully utilized in earlier planetary missions. This research is aimed at addressing the problems associated with designing airbag systems that are sturdy enough to withstand landing impacts during small-scale lunar missions.



RELEVANT LITERATURE SUMMARY

The design of this airbag system for landing is inspired by previous airbag-based landing systems deployed on planetary explorations like the Mars Pathfinder and the Mars Exploration Rover mission. Previous designs have shown that airbags are effective in reducing landing loads on the planet's surface. In these systems, the lander, typically configured in a tetrahedral geometry, was enclosed within airbags that absorbed kinetic energy through compression and controlled venting. The governing physics of these systems includes force balance equations of the form $M\ddot{z} = (P - P_{\text{atm}})A - Mg$, as well as energy conservation principles where the initial kinetic energy of the system is dissipated through gas compression and fabric deformation. Adiabatic compression equations like $PV^\gamma = \text{constant}$ are also applied for calculating airbag pressure dynamics. These studies highlight that while airbags significantly reduce impact loads, the structural system must still be designed to withstand residual forces, particularly during peak deceleration and multiple bounce events.

RELEVANT LITERATURE SUMMARY

Parameter	Luna 9 (USSR, 1966)	Mars Pathfinder (NASA, 1997)	MER Spirit / Opportunity (NASA, 2004)
Destination	Moon (Oceanus Procellarum)	Mars (Ares Vallis)	Mars (Gusev / Meridiani)
Total launch mass	1,538 kg	~890 kg	1,062 kg
Lander / shell mass	~1,439 kg (bus+petals)	~360 kg (landed)	365 kg (lander petal structure)
Rover / payload mass	99 kg (ALS capsule)	10.6 kg (Sojourner)	174 kg (Spirit/Opportunity)
Airbag system mass	~15–20 kg (est.)	98 kg (bags + inflators)	~100 kg (bags + inflators)
Airbag type	Non-vented, ejected from bus	Non-vented, 4 bags × 6 lobes	Vented, 4 bags × 6 lobes
Impact velocity	22 km/h (6.1 m/s)	18 m/s (64 km/h)	~14 m/s vertical, ~12 m/s horizontal
Peak decel (g)	~5–10 g (est.)	~40 g (bouncing)	~40 g (bouncing) - (392 m/s ²)
Airbag fabric	Unknown (Soviet-era)	Vectran (Kevlar-class)	Vectran
Bounces	Several (rolled to rest)	15+ (up to 15 m high)	Multiple (rolled to rest)
Lander structure	Bus + 4-petal capsule	Tetrahedral, 4 petals	Tetrahedral, 4 petals (CFRP)
Power	Silver-zinc battery (~100 W)	Solar + battery (180 W peak)	Solar panels (140 W peak)
Comms	Direct to Earth (Jodrell Bank)	UHF direct + relay	UHF via orbiters

Table 1 - Airbag System Missions

METHODS

The methodology adopted in this work is based on a physics-based analytical modelling approach. In the first step, the energy balance technique was used to model the landing process, assuming that the kinetic energy of the system before impact will be completely absorbed during airbag compression. This leads to the expression $(1/2)MV^2 = F \cdot \Delta x$, from which the impact force was calculated as $F = MV^2/(2\Delta x)$. The force exerted was converted into an effective pressure by means of the relation $P = F/A$, where A represents the contact area.

For structural analysis, two structures were considered: pyramidal (tetrahedral) structure and cube-shaped structure. The pyramidal structure was modeled as an axial loading problem, whereby loads act on the edges and the structural elements, hence giving rise to a membrane stress condition. Conversely, the cube-shaped structure was modeled as a series of plane sections undergoing bending, hence increasing the likelihood of yielding and buckling.

METHODS

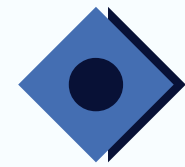
Material properties were considered using the allowable stress concept. This is obtained as $\sigma_{\text{allow}} = \text{FOS} \cdot \sigma_y$, where σ_y is the yield stress of the material, and FOS is the factor of safety. This integrated approach allowed for the derivation of thickness as a function of load, geometry, and material properties.

Material	Density (kg/m ³)	Young's Modulus (GPa)	Yield Strength (MPa)	Allowable Stress (MPa, FOS=2)
Aluminum 6061-T6	2700	69	270	135
Aluminum Honeycomb (thickness 1-5 mm)	80	0.95	2.1	1.05
Steel (mild)	7850	200	250	125

Table 2 - Material comparison

OUTCOMES

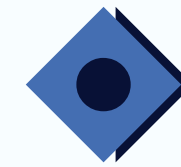
Based on the analysis presented earlier, explicit formulas can be derived for the structural thickness needed in each case:



For the tetrahedral structure, the thickness varies linearly with applied load and is given by -

$$t = MV^2 \cdot FOS / (2\Delta x \cdot L \cdot \sigma_y)$$

This implies high efficiency of such structures due to axial loads transmission.



For the cubic structure, the thickness is governed by the bending behavior and follows -

$$t = \sqrt{(Ca^2 \cdot FOS \cdot MV^2) / (\sigma_y \cdot 2\Delta x A)}$$
 where C is a geometric constant.

A key outcome of the study is that buckling is the dominating factor affecting the structure thickness and should be considered in addition to stress. Numerical evaluation for the given mission parameters shows that impact forces are on the order of hundreds of Newtons, with resulting pressures in the kilopascal range. The tetrahedral structure consistently requires lower thickness compared to the cubic configuration, making it more mass-efficient.

OUTCOMES

Moreover, material comparison revealed that using more advanced aluminum materials as well as composite sandwich structures, the thickness and total weight can be effectively reduced. Honeycomb sandwich structure has a high stiffness-to-weight ratio and is widely used in aerospace applications.

The key takeaway from this analysis is that structural geometry plays a critical role in determining load distribution and required thickness, and that buckling considerations are essential for accurate design.

Mass	16 kg lander (4 plates * 4 kg) + 10 kg (rover and other subsystems)
Velocity	7.2 m/s for 40g impact
FOS	1.5
Δx	40 cm
L	85 cm (triangular panel length)
σ_y	around 270 MPa
Thickness (t)	

Table 3 - Thickness analysis

COMPARISONS

Parameter	Tetrahedral Frame	Cubic Frame	Hexagonal Frame
Basic Geometry	4 triangular faces	6 square faces	6+ polygonal faces
Primary Load Type	Axial (truss-like)	Bending (plate)	Bending + partial load sharing
Structural Efficiency	Very High	Moderate	Better than cube
Thickness Requirement	Lowest	Highest	Medium
Stress Distribution	Uniform, well distributed	Concentrated in panels	More distributed than cube
Buckling Susceptibility	Very low	High	Moderate
Energy Absorption (Impact)	Excellent	Poor–Moderate	Moderate–Good
Mass Efficiency	Best (least material needed)	Worst	Better than cube
Internal Volume Utilization	Poor	Excellent	Good
Ease of Packaging Subsystems	Difficult	Very easy	Easy
Manufacturability	Complex	Simple	Moderate
Integration with Airbags	Proven (used in Mars Pathfinder)	Less common	Conceptual / modern designs
Failure Mode	Member failure (rare)	Panel bending + buckling	Mixed mode

Table 4 - Frame Geometry comparison

COMPARISONS

Criteria	Honeycomb (Sandwich Panel)	Flat (Solid Panel)
Thickness Requirement	Lower face sheet thickness; overall panel thicker due to core height	Requires higher thickness to achieve same stiffness
Stiffness-to-Weight Ratio	Very high (due to core separation increasing moment of inertia)	Low to moderate
Load Carrying Mechanism	Face sheets carry bending; core carries shear	Entire thickness carries bending
Crushing Behavior	Core undergoes progressive crushing → energy absorption	Sudden failure (plastic deformation or fracture)
Energy Absorption	High (ideal for impact/landing systems)	Low to moderate
Buckling Resistance	Excellent (core prevents face wrinkling)	Poor; prone to panel buckling
Mass Efficiency	Very high (lightweight core)	Low (more material required)
Failure Mode	Core crushing + face sheet wrinkling (gradual)	Yielding or buckling (sudden)
Suitability for Airbag Lander	Highly suitable (absorbs residual impact loads)	Less suitable (higher stress transmission)

Table 5 - Panel comparison

CAD AND SYSTEM REPRESENTATION

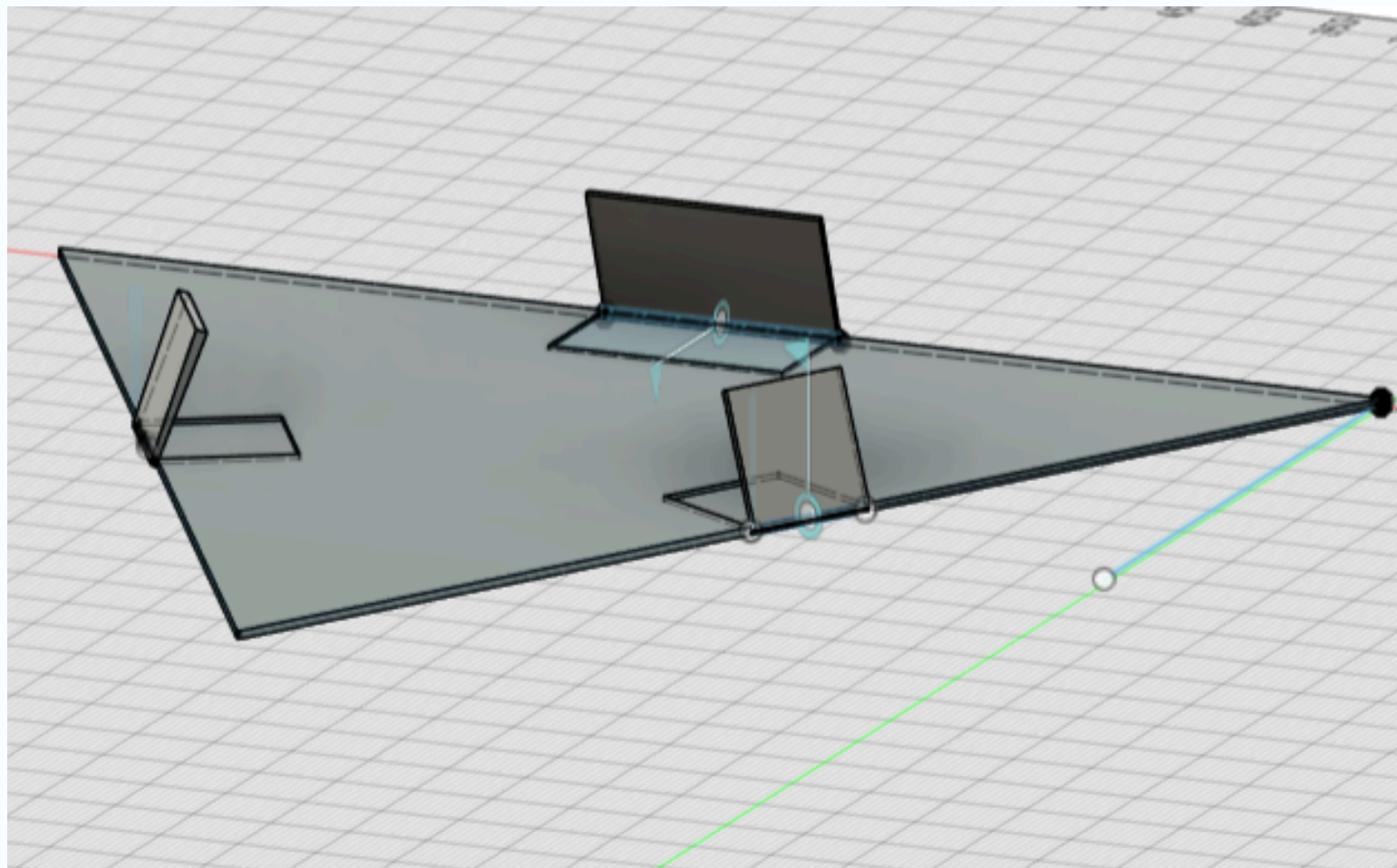


Figure 1 - base panel with hinges

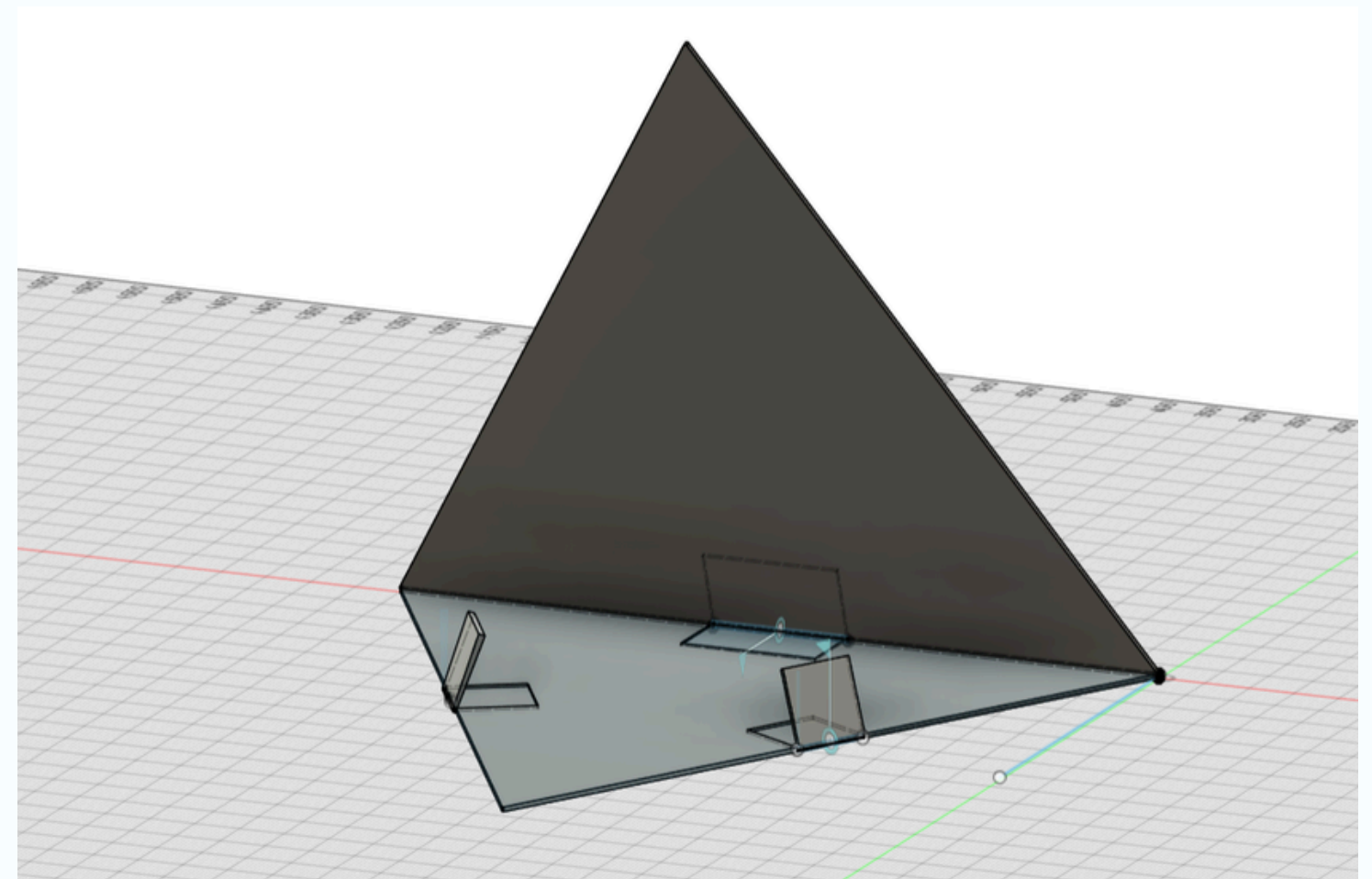


Figure 2 - 2nd petal

CAD AND SYSTEM REPRESENTATION

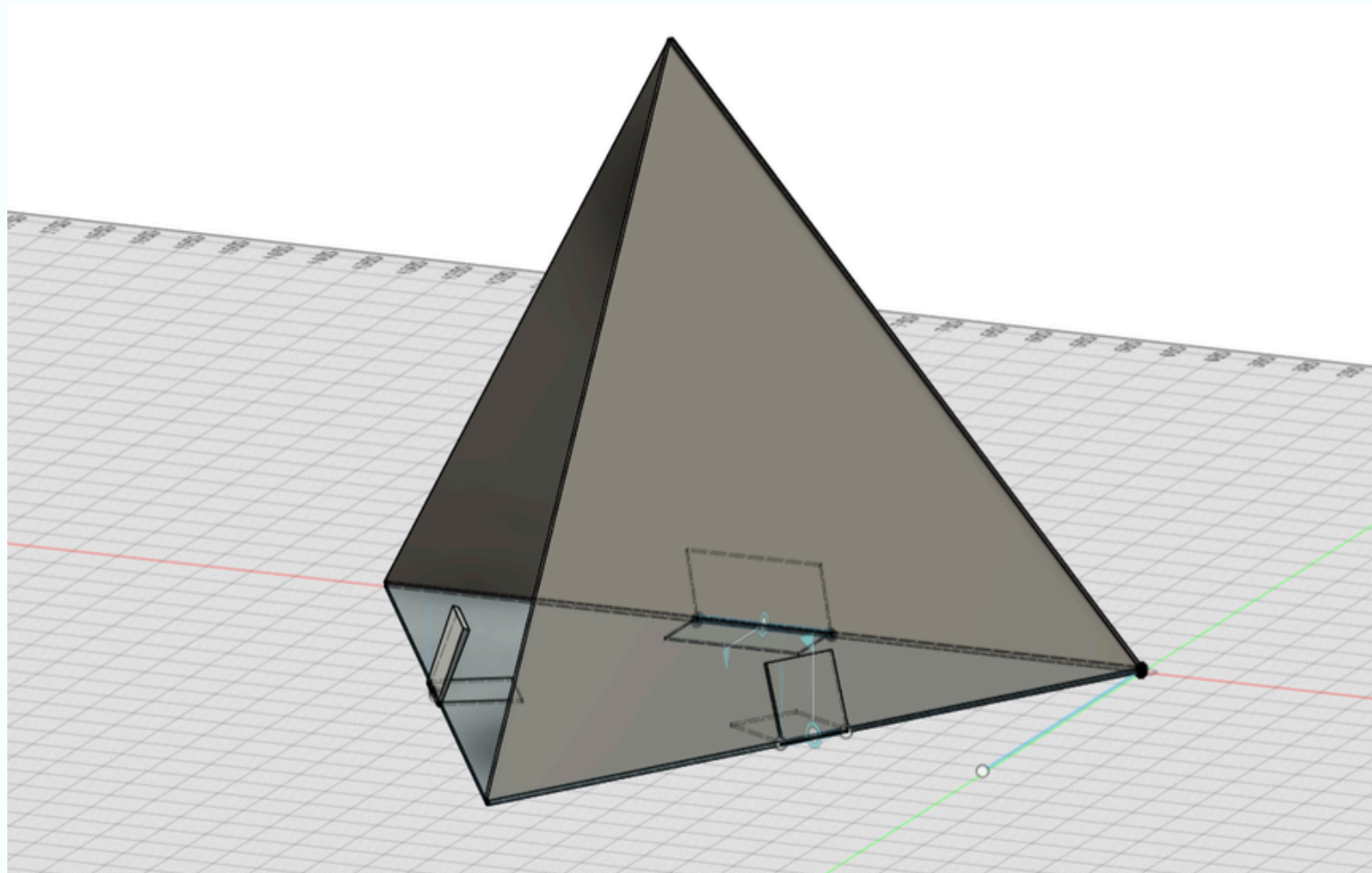


Figure 3 - 3rd petal

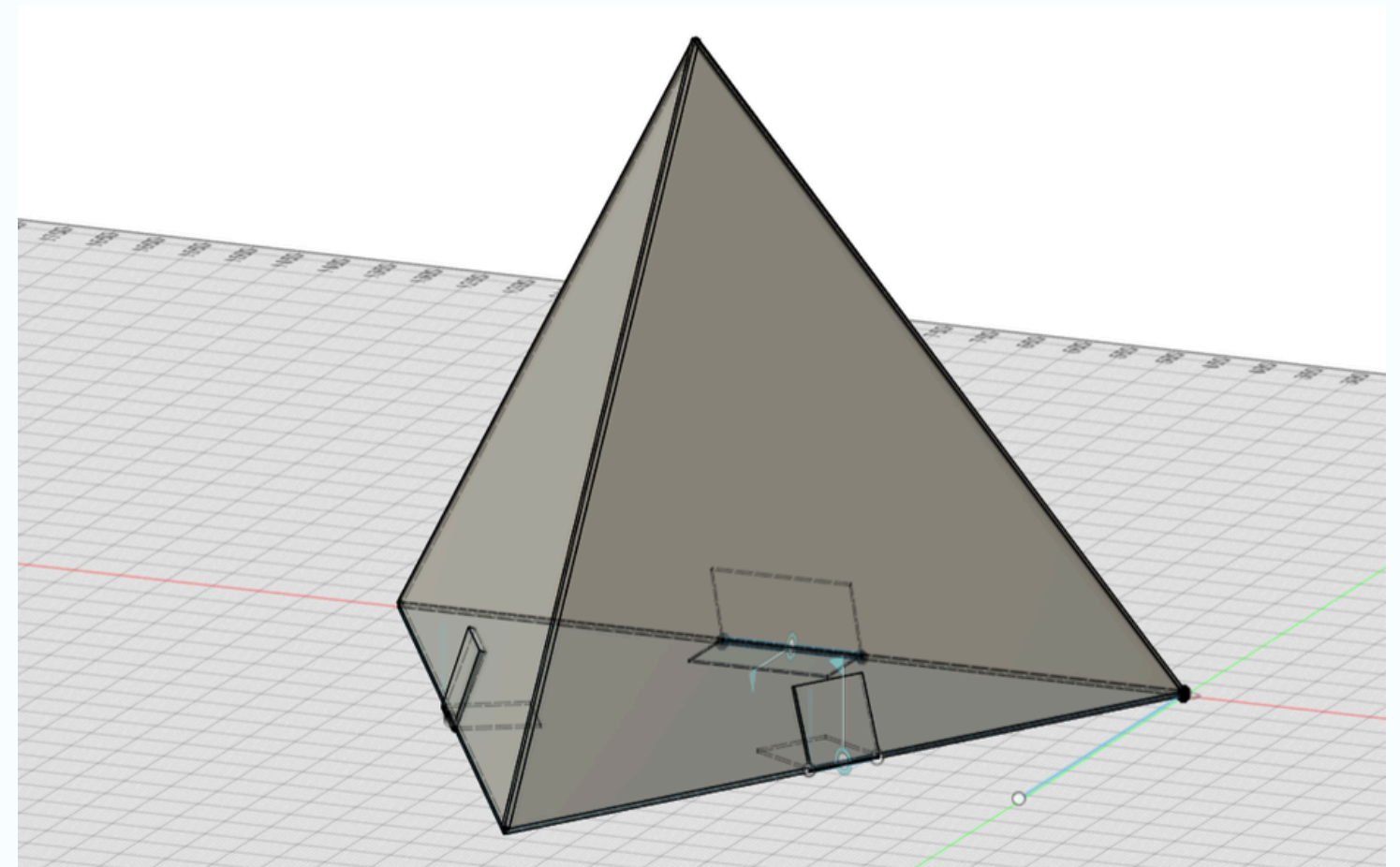


Figure 4 - 4th petal

CAD AND SYSTEM REPRESENTATION - EXAMPLE

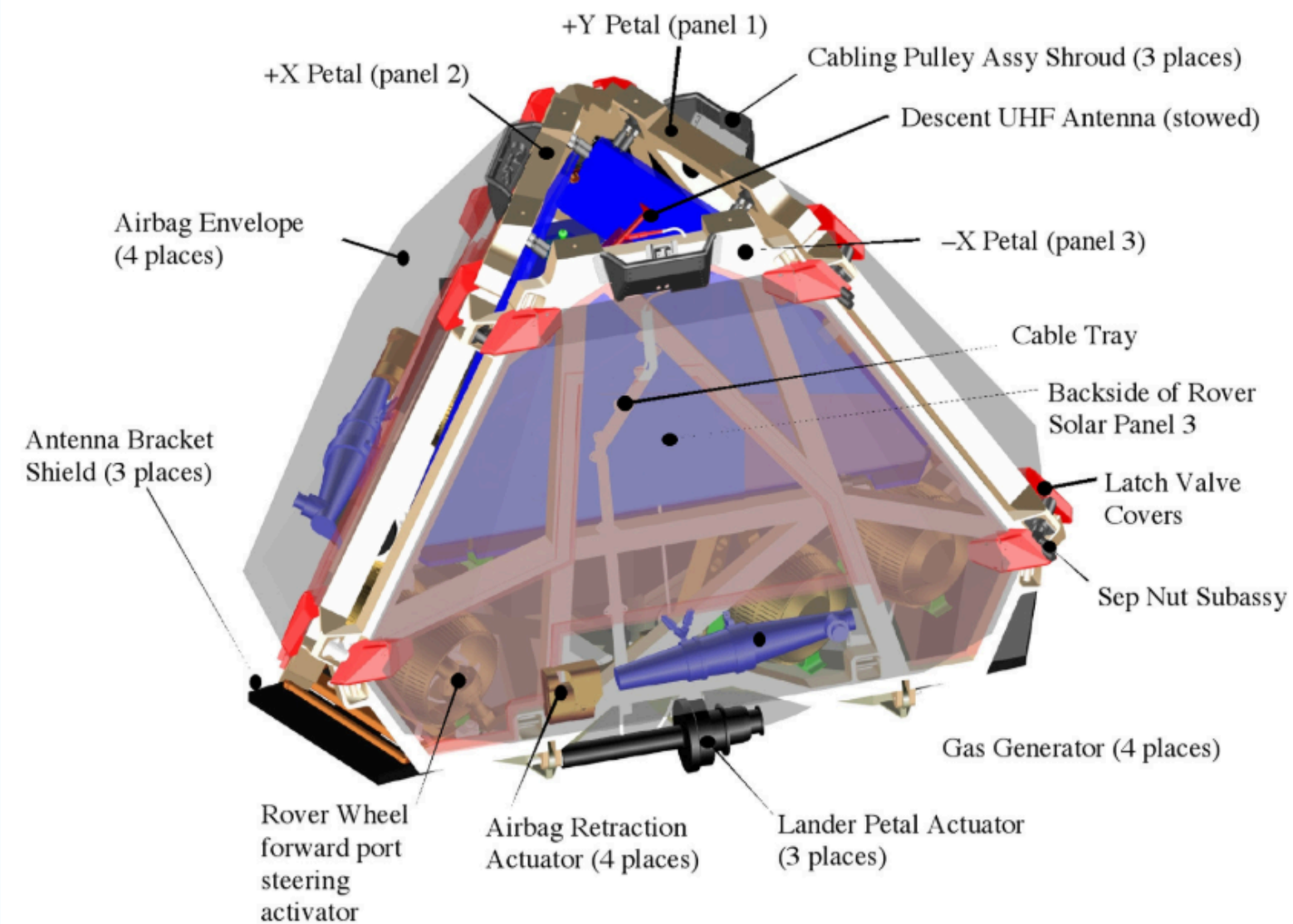


Figure 5 - Stowed Lander

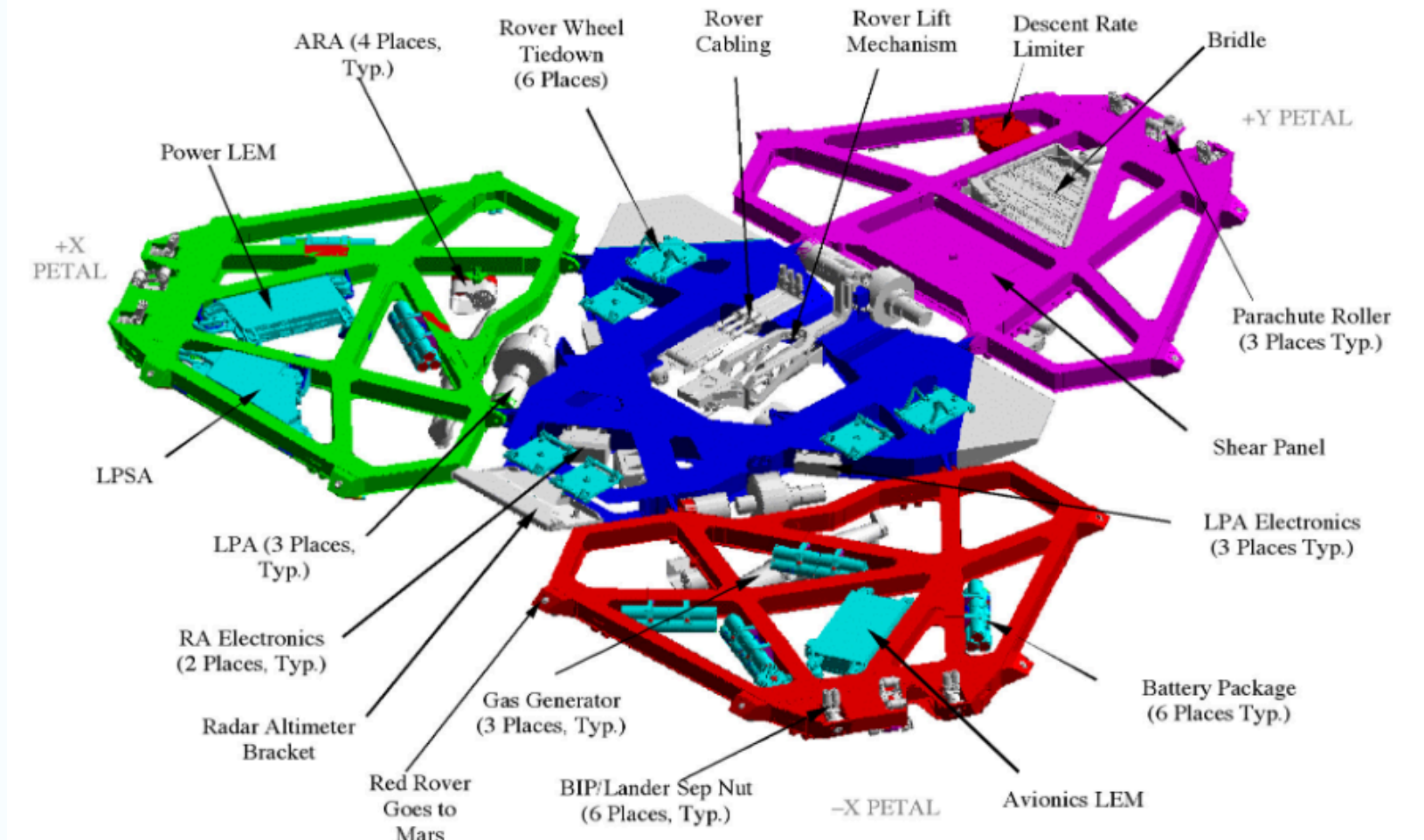
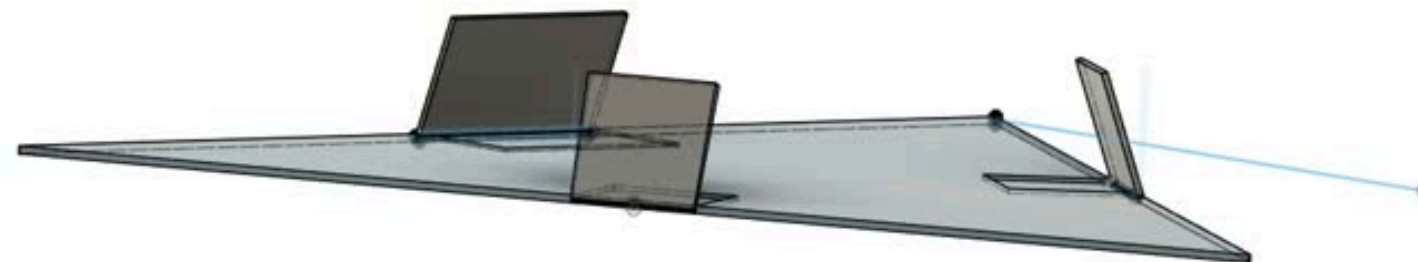


Figure 6 - Deployed Lander

CAD AND SYSTEM REPRESENTATION



Animation video of the lander frame

CONCLUSION

In conclusion, the current study shows that although airbag technology is effective in reducing loads during touchdown, the structural design of the inner shell remains a critical factor in mission success. It is important to balance out all aspects of design to ensure the lightest possible structure while maintaining reliability. The next steps include simulation optimization of the model using computer analysis and designing the technology for more complex structures.

The image features a light blue background with four decorative corner elements. Each corner contains a dark blue square with diagonal lines, and a medium blue square is positioned adjacent to it. The top-left and bottom-right corners have the dark blue square in the foreground, while the top-right and bottom-left corners have the medium blue square in the foreground.

THANK YOU